Highlights

Modelling population distribution among cities on spatial networks using the kinetic theory of multi-agent systems

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- Application of the kinetic theory of multi-agent systems to population distribution on spatial networks.
- Interpretation of cities as interacting agents and interurban links as edges.
- Validation of emigration rules yielding a bimodal lognormal fit for total population distribution and a Pareto fit for the tail.

Modelling population distribution among cities on spatial networks using the kinetic theory of multi-agent systems

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ABSTRACT

This work investigates the population distribution through the kinetic theory of multi-agent systems on spatial networks, interpreting cities both as agents and vertices, and representing interurban links as edges. Various emigration rules have been proposed to regulate the interactions between agents, yielding two primary results: the bimodal lognormal fit of the total population distribution among cities and the Pareto fit of the tail. Overall, the model provides results consistent with the actual distribution derived from ISTAT data of Sardinia; furthermore, the bimodal lognormal fit generally follows the distribution tail more accurately than the Pareto one, although the two coincide in the first half.

1. Introduction

The study of population distribution among cities is a phenomenon that has been sporadically analyzed in the past. The first was Auerbach [1] who, at the beginning of the 20th century, noted a characteristic later formalized by Zipf[1] [20] almost fifty years later, albeit initially in a linguistic context; indeed, Zipf's [empirical] law states

$$f \propto \frac{1}{r} \quad \text{where} \begin{cases} f \text{ is the frequency of the word,} \\ r \text{ is its rank in the classification;} \end{cases} \tag{1}$$

to give a concrete example, if the tenth most frequent word appears 2000 times, then there exists a constant C such that $2000 \approx C/10$. The same law described in (1) can be found in many other contexts, including the distribution of population among cities if, instead of f, one writes the population s [20, Fig. 9-2 p. 375]; however, this is true only when $s \gg 1$, namely in the tail of the distribution, contrary to Zipf's original linguistic context where (1) always holds.

More generally, Zipf's law can be described by a Pareto distribution: let $S \in \mathbb{R}_+$ be the random variable related to city size and let $f_S(s)$ be its probability density function[2], which allows interpreting $f_S(s)\mathrm{d}s$ as the number of cities with $s \in [s, s + \mathrm{d}s]$; then S is said to follow a Pareto distribution in the tail if it satisfies

$$\mathcal{R}(s) \equiv \int_s^{+\infty} f_S(z)\mathrm{d}z \approx \frac{c}{s^\beta} \quad \text{if } s \gg 1, \tag{2}$$

for a certain constant $c \in \mathbb{R}_+$ and with $\mathcal{R}(s)$ equal to the complementary cumulative distribution function (CCDF) of S: it represents the number of cities with a population greater than or equal to s and, if the Pareto index $\beta \in \mathbb{R}$ were unitary, it is analogous to the rank r in (1).

More recently [8, 9], it has been discovered that outside the tail, $f_S(s)$ is well fitted by the lognormal distribution with probability density

$$f_X(x) = \frac{1}{\ln(10)} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \tag{3}$$

where $X \sim \mathrm{Lognormal}(\mu, \sigma)$; as the name suggests, the peculiarity of this distribution is that $Y = \log X \sim \mathcal{N}(\mu, \sigma)$ holds, where $\mathcal{N}(\mu, \sigma)$ indicates the Gaussian distribution with mean μ and standard deviation σ.

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[1]Actually, as Zipf himself admits [20, p. 546], he was not the first to discover such a law; in any case, he popularized it to such an extent that today it is known by his name.

[2]For $f_S(s)$ to exist, one must strictly also assume that S is absolutely continuous, but the details are not necessary here.

Furthermore, Gualandi et al. [9, 10] realized that the entire distribution is well fitted by a bimodal lognormal (here also denoted as bilognormal), namely a convex combination of (3)

$$f_S(s) \approx f_X(s) = \xi f_{X_1}(s) + (1-\xi) f_{X_2}(s), \tag{4}$$

where $\xi \in (0,1)$, $X_1, X_2 \sim \text{Lognormal}(\mu, \sigma)$ and $X \sim \text{BiLognormal}(\xi, \mu_1, \sigma_1, \mu_2, \sigma_2)$.

Therefore, the objective of this paper is as follows: we want to simulate through the Kinetic Theory of Multi-Agent Systems (KTMAS) the interactions between cities on a spatial graph, attempting to reproduce a population distribution with characteristics analogous to those actually measured (mainly bilognormal body and Pareto tail) in the Sardinia region of Italy. Once this goal is achieved, the migration phenomenon is interpreted through the proposed emigration laws, which are the basis of the interactions between cities.

Moreover, it is emphasized that in this work, cities are considered as agents characterized by their population, which interact through the underlying structure of a [spatial] graph; this description, except for the network, is not unlike the classical one of gas molecules characterized by their velocity and position. In fact, in the literature, at least to the authors' knowledge, there are no articles dealing simultaneously with the KTMAS, population distribution among cities, and graphs, despite the natural representation of these on a network; more in detail:

- some either apply the KTMAS to cities without considering a natural underlying topology [10][3] or, conversely, consider the graph structure without applying it [4];
- others view nodes more as places where agents live and interact, thus creating a model described by a system of large, coupled Boltzmann equations [14];
- finally, the article closest to the current paper applies its theory in the context of social networks [16].

This gap is probably attributable to the fact that developments in graph theory applied to the KTMAS have been quite recent and primarily focused on the node-place perspective rather than node-agent. Therefore, the most innovative aspect of this work is showing that the second perspective is as valid as the first, despite being less common; otherwise, this paper should be considered as a formal applicative exploration with few analytical in-depth studies.

Overall, the manuscript is divided as follows: the second section will briefly and superficially address some notions of the KTMAS mediated by graphs; then the third will showcase the numerical results for various emigration rules for the Sardinia region; finally, some conclusions are drawn with potential future developments.

2. Notes of the KTMAS on graphs

In this section some hypotheses about the type of graph used and the two main Boltzmann type equations on networks are briefly discussed. For the general theory of the KTMAS see [15, 17].

Before continuing, two recurrent definitions are the following ones: $\mathbb{R}_+ \equiv \{x \in \mathbb{R} : x \geq 0\} \subset \mathbb{R}$ is the subset of the real non-negative numbers, while $\mathbb{N}_+ \equiv \{n \in \mathbb{N} : n \geq 1\} \subset \mathbb{N}$ is the subset of the natural positive numbers.

2.1. Networks and cities

First, it's important to clearly define what a graph is:

Definition 1 (Graph). A graph is a set $\mathcal{G} = (\mathcal{I}, \mathcal{E})$ formed by the pair of the set of node indices $\mathcal{I} \subseteq \mathbb{N}_+$ and the set of edges $\mathcal{E}$, namely pairs of indices of $\mathcal{I}$: two nodes $i, j \in \mathcal{I}$ are connected if and only if $(i, j) \in \mathcal{E} \subseteq \mathbb{N}_+ \times \mathbb{N}_+$.

On the other hand, other related concepts are henceforth taken for granted though they can be found in any reference about graph theory like [3, 18].

The next problem is how to exactly interpret a link between two nodes, i.e. cities, given the multitude of possible urban connections. In fact, there are many ways to represent cities by means of graphs: the first are at the *intraurban* level, either considering streets as edges and their intersections as nodes [3, § 3.1.3.1 p. 17], or representing transport networks (tram, bus, and metro) [3, § 3.2.1 p. 22]; others are more properly at the *interurban* level, taking individually into consideration various transport networks (railway [3, § 3.1.3.2 p. 17], naval [3, § 3.1.4 p. 19], air [3, § 3.1.2 p. 13], etc.). However, from what was stated in § 1, it is clear that in order to predict the population distribution among cities, the following observations hold true:

[3] In particular [10], despite being a very interesting article that reaches equally important conclusions, excessively abstracts the interactions between cities and fails to define with sufficient clarity what is meant by *neighbourhoods*.

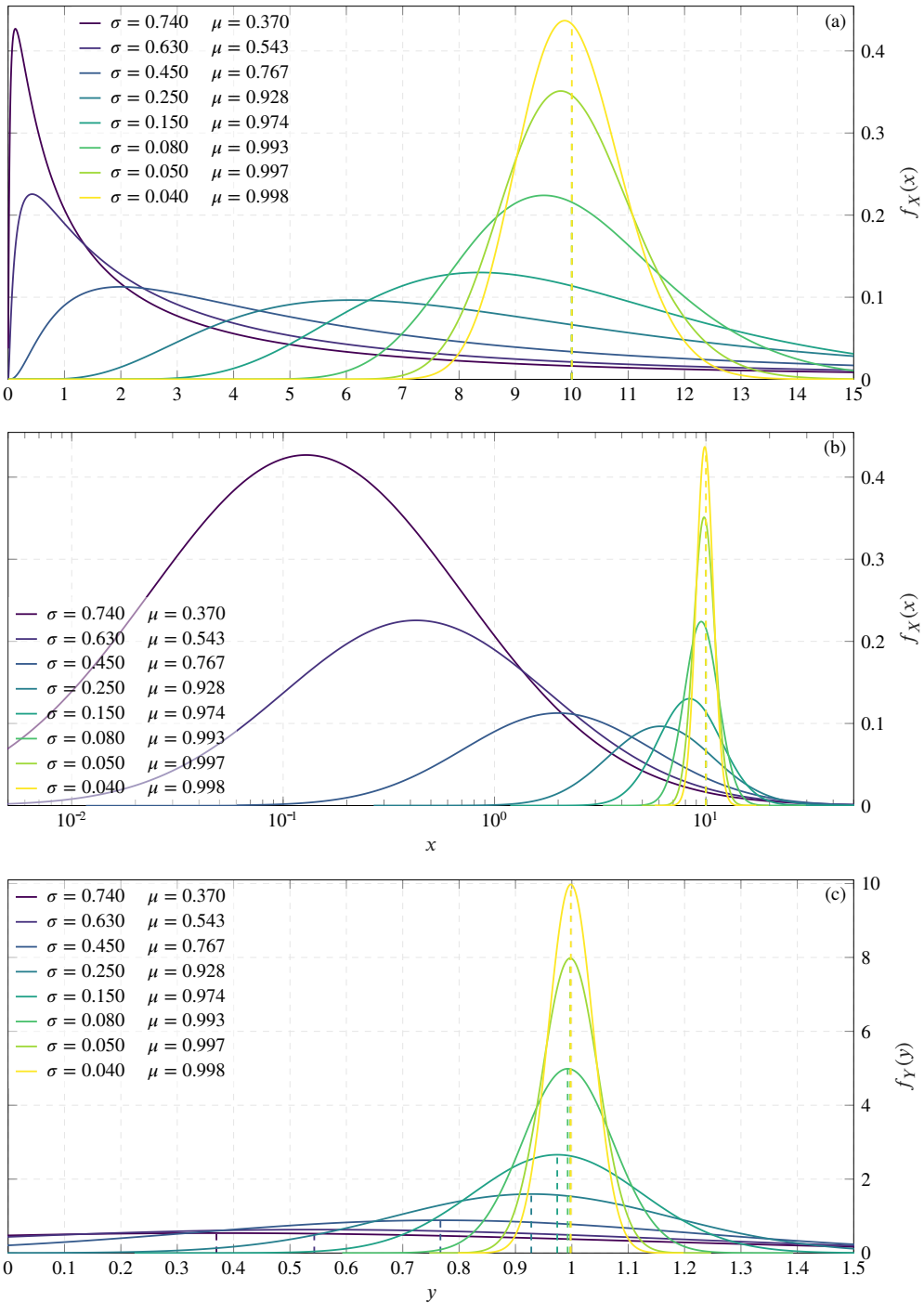


Figure 1: Function $X \sim \text{Lognormal}(\mu, \sigma)$ for varying parameters, but with equal expected value $\mathbb{E}[X] = 10^{\mu+\sigma^2 \ln(10)/2} = 10$, in linear scale (a) and logarithmic scale (b); instead in (c) the same distributions are reported in logarithmic space $y = \log(x)$. The dashed lines represent the expected values of each distribution.

1. every intraurban representation must be discarded because its movements are limited to a single city;
2. on the other hand, all the interurban ones must be considered simultaneously and not individually, since each person can choose different modes of transport to move around.

This is why the correct network used in the simulation is the one related to commuting movements [3, § 3.1.3.3 p. 18; 6] between cities, which holistically shows all their possible connections regardless of the chosen transportation mode; furthermore, it shows realistic connections associated with daily rather than extraordinary movements (holidays, medical visits, etc.). Once the representation is fixed, it is necessary to understand the type of graph we are dealing with. Yet, even here, the answer is immediate after two further notes:

1. a city can interact with another without the latter interacting with the former;
2. assuming a city can interact with another, then the opposite is always possible.

Therefore the sense of direction must be considered and the potential interactions are symmetric, that is the graph in question is directed but with an undirected structure:

Hypotesis 1. The graph $\mathcal{G}$ is directed and symmetric, hence the adjacency matrix $\mathbf{A}$ is symmetric.

Furthermore, the following fundamental hypothesis holds in this discussion:

Hypotesis 2. The graph $\mathcal{G}$ is static: $\mathcal{I}$ and $\mathcal{E}$ are constant over time.

In other words, we are only interested in predicting how the population distributes itself with respect to a topology established *a priori*; naturally, this is a strong simplification since in reality the topology, especially the interurban connections in $\mathcal{E}$, has clearly co-evolved along with the development of the cities themselves. Finally, it is observed that this type of graph is spatial [3, 6], meaning its nodes occupy a point in Euclidean space, even though, at the moment, this is a somewhat formal characteristic; however, it helps to keep it in mind when interpreting certain emigration rules in § 3.2. Moreover, contrary to what one might think at first glance, this graph is not scale-free [2] precisely because of its spatial nature [5, 3]; this does not rule out the existence of nodes that are more central than others, but only implies that the maximum degree of a node is upper-bounded due to physical constraints.

2.2. Exact Boltzmann-type equation

The KTMAS is centred around agents interacting to one another in so changing their microstate $\mathbf{x}_t \in \mathbb{R}^n$; however, usually the precise knowledge of a single agent's microstate is neither useful nor necessary, hence why a stochastic approach is taken by denoting $\mathbf{X}_t \in \mathbb{R}^n$ as the random variable of the microstate at time t; finally, the sequence in time of microstates $\{X_t\}_{t\in\mathbb{R}_+}$ is called a stochastic process. To derive the equation for the [probability] density function $f(\mathbf{x}, t)$ four assumptions are made [15]:

Hypotesis 3. Agents are characterized by the following hypotheses:

B1 the statistical distribution f of microstates is spatially uniform;
B2 non-binary interactions between agents are negligible;
B3 the agents are indistinguishable, and each represents the ensemble;
B4 two interacting agents are statistically independent.

Thus a inconsistency arises from B3 if agents are confused with graph nodes: the former are indistinguishable, but the underlying network makes the latter distinct due to the indices $\mathcal{I}$; to solve this apparent issue it's sufficient to consider the index as a random variable $I \in \mathcal{I}$, resulting in the stochastic process $\{\mathbf{X}_t\} \equiv \{(I, S_t)\}_{t\in\mathbb{R}_+}$ where $S_t \in \mathbb{R}_+$ is the random variable related to the population of the representative agent at time t. Of course by definition S_t is the only component of $\mathbf{X}_t$ to vary over time due to Hypothesis 2.

Remark 1 (On the numerical nature of the population). It is natural that fractions of people do not exist and therefore strictly $S_t \in \mathbb{N}$, but the most realistic choice is not always the easiest modelling-wise. At any rate, if necessary, it's possible to gauge the actual population by rounding it down to the nearest natural number.

Reasoning in this way, one can overall statistically describe the microscopic state $\mathbf{X}_t \equiv (I, S_t)$ of the representative agent with the joint density $f : \mathcal{I} \times \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$, which is discrete in $i \in \mathcal{I}$ and continuous in $s \in \mathbb{R}_+$, equal to

$$f(i, s, t) \equiv \frac{1}{N} \sum_{j \in \mathcal{I}} f_j(s, t) \otimes \delta(i - j), \tag{5}$$

where $N \equiv |\mathcal{I}|$ is the total number of agents in the graph, while $\delta(\cdot)$ denotes the Dirac delta centered at the origin; on the other hand, $f_i : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$ is the density of S_t relative to the i-th agent (Figure 2).

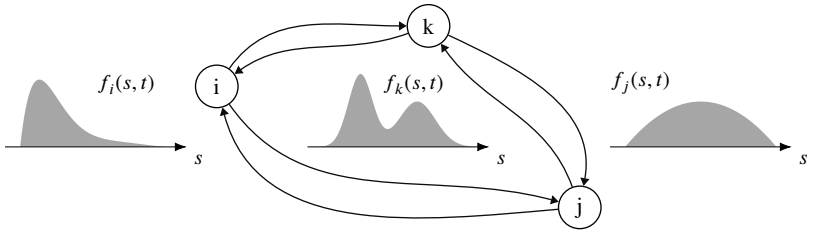


Figure 2: Graphical representation of the KTMAS on graphs.

In this way, f is effectively indistinguishable with respect to all agents and integrates over its domain to one:

$$\begin{aligned}\mathbb{P}_t(\mathcal{I} \times \mathbb{R}_+) &= \int_{\mathcal{I}} \int_{\mathbb{R}_+} f(i, s, t)\mathrm{d}s\mathrm{d}i = \frac{1}{N} \sum_{i \in \mathcal{I}} \int_{\mathcal{I}} \int_{\mathbb{R}_+} f_i(s, t)\mathrm{d}s \otimes \delta(i - j)\mathrm{d}i \\ &= \frac{1}{N} \sum_{i \in \mathcal{I}} \int_{\mathcal{I}} \delta(i - j)\mathrm{d}i = \frac{1}{N} \sum_{i \in \mathcal{I}} 1 = \frac{N}{N} = 1, \quad \forall t \geq 0,\end{aligned}$$

allowing to preserve the natural distinguishability of the agents as well, induced by the underlying topology through the individual densities f_i.

When two agents interact they alter their respective microstate following the following general scheme

$$[\mathrm{RI}]_S \begin{cases} S_t' = \psi(S_t, I, S_t^*, I_*), \\ S_t^{*\prime} = \psi_*(S_t, I, S_t^*, I_*). \end{cases} \quad \text{where the functions} \quad \begin{aligned} \psi &: \mathbb{R}_+ \times \mathbb{N}_+ \times \mathbb{R}_+ \times \mathbb{N}_+ \to \mathbb{R}_+, \\ \psi_* &: \mathbb{R}_+ \times \mathbb{N}_+ \times \mathbb{R}_+ \times \mathbb{N}_+ \to \mathbb{R}_+ \end{aligned}$$

describe the post-interaction microstate; however, it's not certain that two randomly picked agents can interact: this can be taken into account with the Bernoulli

$$\Theta \sim \mathrm{Bernoulli}(A(I, I_*)\Delta t), \text{ with } A(I, I_*) \equiv a_{I,I_*} = \begin{cases} 1 & \text{if } (I, I_*) \in \mathcal{E}, \\ 0 & \text{otherwise,} \end{cases} \qquad [\mathrm{Ber}]_S^{\mathbf{A}}$$

where $A : \mathcal{I} \times \mathcal{I} \to \{0, 1\}$ is a function that associates each pair of indices (I, I_*) with the corresponding element of the adjacency matrix; defined in this way, $[\mathrm{Ber}]_S^{\mathbf{A}}$ inhibits *a priori* the interaction if the two agents are not connected.

Remark 2. By definition of the Bernoulli distribution, it is necessary to impose $A(I, I^*)\Delta t \leq 1$, but given that $A(I, I^*) \in \{0, 1\}$ this translates into the natural condition that $\Delta t \leq 1$.

Combining $[\mathrm{Ber}]_S^{\mathbf{A}}$ and $[\mathrm{RI}]_S$, the following algorithm is obtained

$$[\mathrm{AR}]_S \begin{cases} S_{t+\Delta t} = (1 - \Theta)S_t + \Theta S_t', \\ S_{t+\Delta t}^* = (1 - \Theta)S_t^* + \Theta S_t^{*\prime}, \end{cases}$$

consistently with the fact that in an urban context, if a city-node interacts with another, both must vary their state.

Remark 3. The acronym $[\mathrm{AR}]_S$ stands for Action-Reaction and it assumes that every interaction (action), i.e., $\Theta = 1$, necessarily modifies the states of both involved agents (reaction); in certain contexts [16, (2.5) p. 5] this rule can be altered such that only the interacting agent is modified, leading to Action-Action interactions.

Let now $\Phi : \mathcal{I} \times \mathbb{R}_+ \to \mathbb{R}$ be an arbitrary observable quantity, then by averaging it while evaluating $[\mathrm{AR}]_S$ it can be shown [16] to lead to the homogeneous Boltzmann-type equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{I}}\int_{\mathbb{R}_+}\Phi f\,\mathrm{d}s\mathrm{d}i = \int_{\mathcal{I}^2}\int_{\mathbb{R}_+^2} A(i,i_*)\frac{\Phi'+\Phi'_*-\Phi-\Phi_*}{2} f f_*\,\mathrm{d}s\mathrm{d}s_*\mathrm{d}i\mathrm{d}i_*, \tag{$[\mathrm{EtB}]_S^{\mathbf{A}}$}$$

where an abbreviated notation has been used in which the arguments are omitted and the asterisks as well as the primes change their type: for instance $\Phi'_* \equiv \Phi(i_*, s'_*)$ while $f_* \equiv f(i_*, s_*, t)$, and so forth.

2.3. Approximated Boltzmann-type equation

Recently an approximation of $[\mathrm{EtB}]_S^{\mathbf{A}}$ has been proposed [16] based on the adjacency matrix $\mathbf{A}$:

Definition 2 (Degree matrix B). Let $\mathbf{B} \in \mathbb{R}^{N\times N}$ be the rank-one matrix

$$\mathbf{B} \equiv \frac{\mathbf{k}^+(\mathbf{k}^-)^\top}{D_N} \quad \text{with} \quad \begin{cases} \mathbf{k}^+ \equiv (k_1^+, k_2^+, \dots, k_N^+)^\top = \mathbf{A}\mathbf{1} \in \mathbb{R}^N, \\ \mathbf{k}^- \equiv (k_1^-, k_2^-, \dots, k_N^-)^\top = \mathbf{A}^\top\mathbf{1} \in \mathbb{R}^N, \end{cases}$$

being the out-degrees and in-degrees respectively, and the constant $D_N \equiv \mathbf{1}^\top\mathbf{A}\mathbf{1}$. Then $\mathbf{B}$ approximates adjacency matrix $\mathbf{A}$ since by definition they share the same in- and out-degrees:

$$\left.\begin{aligned} \mathbf{B}\mathbf{1} &= \frac{1}{D_N}\mathbf{k}^+(\mathbf{k}^-)^\top\mathbf{1} = \mathbf{k}^+\frac{D_N}{D_N} = \mathbf{k}^+ = \mathbf{A}\mathbf{1} \\ \mathbf{B}^\top\mathbf{1} &= \frac{1}{D_N}\mathbf{k}^-(\mathbf{k}^+)^\top\mathbf{1} = \mathbf{k}^-\frac{D_N}{D_N} = \mathbf{k}^- = \mathbf{A}^\top\mathbf{1} \end{aligned}\right\} \implies \mathbf{B} \approx \mathbf{A}. \tag{6}$$

Furthermore, since $\mathbf{A}$ is symmetric (Hypothesis 1), $\mathbf{k}^+ = \mathbf{k}^- = \mathbf{k}$ holds and therefore $\mathbf{B} = \mathbf{k}\mathbf{k}^\top/D_N$.

By using $\mathbf{B}$ $[\mathrm{EtB}]_S^{\mathbf{A}}$ becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathcal{I}}\int_{\mathbb{R}_+}\Phi g\,\mathrm{d}v\mathrm{d}i = \int_{\mathcal{I}^2}\int_{\mathbb{R}_+^2} B(i,i_*)\frac{\Phi'+\Phi'_*-\Phi-\Phi_*}{2} g g_*\,\mathrm{d}s\mathrm{d}s_*\mathrm{d}i\mathrm{d}i_*, \tag{$[\mathrm{EtB}]_S^{\mathbf{B}}$}$$

in which the density g is used to distinguish it from f though it has its same definition found in (5); $[\mathrm{EtB}]_S^{\mathbf{B}}$ can be shown [16, Theorem 6.1.] to converge in the limit $N \to +\infty$ to

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^1\int_{\mathbb{R}_+}\varphi\hat{d}\,\mathrm{d}s\mathrm{d}\hat{k} = \int_0^1\int_0^1\int_{\mathbb{R}_+^2}\frac{\hat{k}\hat{k}_*}{\bar{D}}\frac{\varphi'+\varphi'_*-\varphi-\varphi_*}{2}\hat{d}\hat{d}^*\,\mathrm{d}s\mathrm{d}s_*\mathrm{d}\hat{k}\mathrm{d}\hat{k}_*. \tag{$[\mathrm{EtB}]_S^{k}$}$$

where

- Φ has been reduced to an arbitrary observable $\varphi : \mathbb{R}_+ \to \mathbb{R}$ function only of the population s;
- $\hat{d}(s,\hat{k},t)$ is the limit of the distribution of normalized degrees $\hat{d}_N : \mathbb{R}_+ \times \hat{\mathcal{K}} \times \mathbb{R}_+ \to \mathbb{R}_+$, where

  $$\hat{\mathcal{K}} \equiv \{k/N \mid k = 0, 1, \dots, N\} \subseteq [0,1];$$

- the constant $\bar{D}$ is defined as $\bar{D} \equiv \int_0^1\int_{\mathbb{R}_+}\hat{k}\hat{d}\,\mathrm{d}s\mathrm{d}\hat{k}$.

The implications of $[\mathrm{EtB}]_S^k$ are profound: approximating $\mathbf{A}$ with $\mathbf{B}$ coincides with losing the distinguishability induced by the graph as the microstate loses the index i in place of the degree k, thereby relaxing the topology (Figure 3) which does not vanish but remains solely as a degree distribution; such a reduction is akin to blurring the connection details among the vertices: one transitions from a specific graph to a class of graphs sharing the same degree distribution. Therefore $[\mathrm{EtB}]_S^k$ is closer the original context of the KTMAS making it a good candidate to approximate $[\mathrm{EtB}]_S^{\mathbf{A}}$.

Remark 4. $[\mathrm{EtB}]_S^k$ justifies why one cannot consider a generic matrix $\mathbf{C}$ approximating $\mathbf{A}$ in any way: it is not guaranteed to relax the topology and lead to an equation analogous to it like $\mathbf{B}$ did.

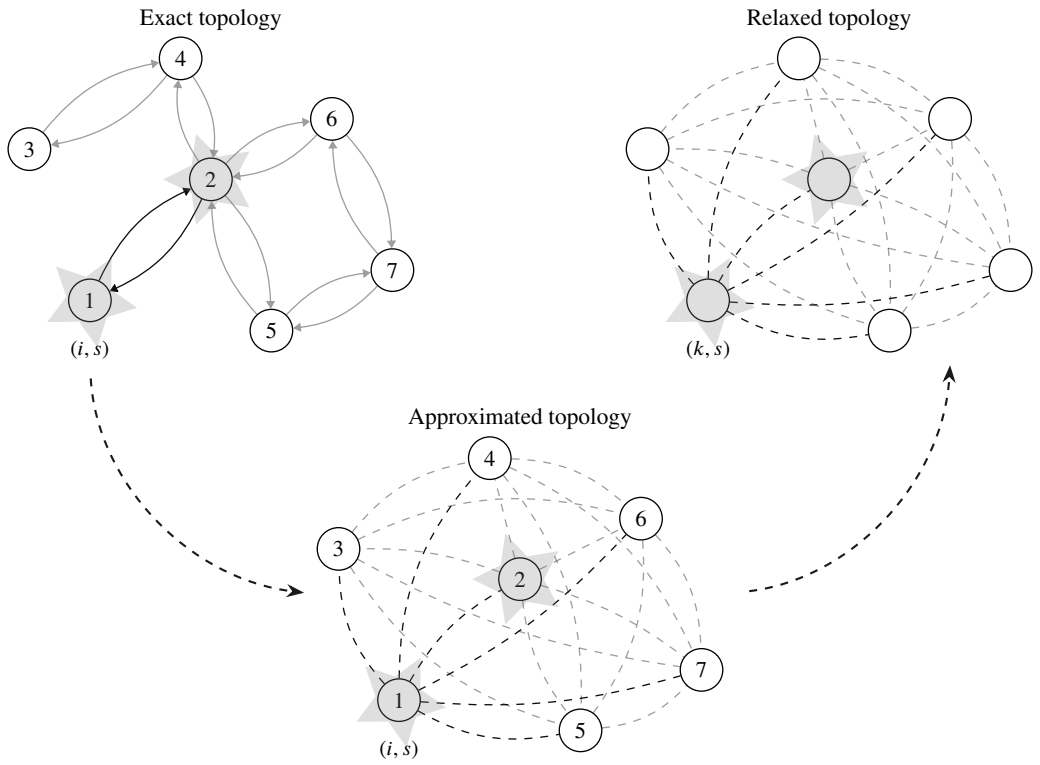


Figure 3: Graphical representation of the KTMAS approximation on graphs; for visual simplicity an undirected graph has been drawn, instead of the more correct directed one (Hypothesis 1), for the approximated and relaxed topology.

2.4. Emigration rules

First, by using realizations of all considered random variables, writing them in lowercase and omitting time dependence—equations, $[\mathrm{RI}]_S$ can be written as follows[4]:

$$[\mathrm{REI}] \begin{cases} \psi(s,i,s_*,i_*) \equiv s\big(1 - E(s,i,s_*,i_*)\big) \\ \psi_*(s,i,s_*,i_*) \equiv s_* + sI(s,i,s_*,i_*) \end{cases}$$

where $E, I : \mathbb{R}_+ \times \mathcal{I} \times \mathbb{R}_+ \times \mathcal{I} \to \mathbb{R}_+$, are the emigration and immigration rates, respectively, whose arguments are implicitly assumed hereafter for brevity like in $[\mathrm{EtB}]_S^{\mathbf{A}}$.

Remark 5. One could also more realistically define $\psi(s,i,s_*,i_*\gamma) \equiv s\big(1 - E(s,i,s_*,i_*) + \gamma\big)$ with γ being the random variable of the variation rate, i.e. the sum of the death and birth rates, representing the natural stochastic fluctuations of the population; [REI] just assumes that $\gamma = 0$: the birth and death rates are always equal.

We further characterize E and I with two hypotheses:

Hypotesis 4 (Positivity constraints). To ensure a physically meaningful (i.e., non-negative) post-interaction state s', it is assumed that E is upper-bounded by the constant $\lambda \equiv \sup E \in (0,1)$ and lower-bounded by the origin:

$$0 \le E \le \lambda < 1, \quad \forall s,\ \forall s_*,\ \forall i,\ \forall i_*.$$

Conversely, the post-interaction state s'_* is not restricted since it is by definition always positive regardless of the immigration rate $I \in \mathbb{R}_+$.

[4]They were partially inspired by [10, (2.1) p. 223].

Algorithm 1: Nanbu-Babovsky type algorithm $[\mathrm{AR}]_S$

Data: $N \in \mathbb{N}_+$, (number of agents), $\Delta t \leq 1$, (time step), σ, (fluctuation), $T > 0$, (final time), P, (initial total population), $\mathbf{A}$ (adjacency matrix) and $\mathbf{B}$ (approximated adjacency matrix);

1 $\mathcal{S}^0 \leftarrow (s_1^0, s_2^0, \dots, s_N^0) \equiv (P/N)\mathbf{1} \in \mathbb{R}_+^N$;
2 **for** $n = 0, 1, 2, \dots, \lfloor T/\Delta t \rfloor - 1$ **do**
3 $p \leftarrow$ independent permutation of $\{1, 2, \dots, N\}$;
4 **for** $c = 1, 2, \dots, \lfloor N/2 \rfloor$ **do** `// number of pairs`
5 $i \leftarrow p(c)$; $r \leftarrow p(\lfloor N/2 \rfloor + c)$;
6 **if** *exact* **then**
7 $\Theta \leftarrow \mathrm{Bernoulli}(A(i, r)\Delta t)$;
8 **else** `// it is approximated`
9 $\Theta \leftarrow \mathrm{Bernoulli}(B(i, r)\Delta t)$;
10 **if** $\Theta = 1$ **then**
11 $E \leftarrow E(s_i^n, i, s_r^n, r)$;
12 $s_i^{n+1} \leftarrow s_i^n(1 - E)$; `// interacting city`
13 $s_r^{n+1} \leftarrow s_r^n + s_i^n E$; `// receiving city`
14 **else**
15 $s_i^{n+1} \leftarrow s_i^n$; $s_r^{n+1} \leftarrow s_r^n$;
16 $\mathcal{S}^{n+1} \leftarrow (s_1^{n+1}, s_2^{n+1}, \dots, s_N^{n+1})$;
17 $\bar{z}_N(s, (n+1)\Delta t) \leftarrow$ histogram of $\mathcal{S}^{n+1}$;

Hypotesis 5 (Conservation of average population). It is postulated that the [REI] conserve the total population on average: $s + s_* = \langle s + s_* \rangle = \langle s' + s'_* \rangle = s - sE + s_* + sI$, which yields $I \equiv E$, a reasonable result since emigration and immigration are relative phenomena (inverting s and s_* would produce the opposite case).

Remark 6. The [REI] are not yet complete since the emigration rule E has not been specified, the choice of which can be seen as the interaction rule itself; this is done later on in § 3.2 when the results are actually discussed.

3. Implementation and simulations

3.1. Monte Carlo method

As is clear from § 2, the main objective is to obtain the density f in $[\mathrm{EtB}]_S^{\mathbf{A}}$ which is, however, quite difficult to handle due to, among others, its integro-differential nature. Because of this, and as usual in the KTMAS, a simpler procedure is adopted: knowing the algorithm $[\mathrm{AR}]_S$, which governs the binary interactions between agents, all multiple collisions can therefore be directly simulated using a Monte Carlo method of the Nanbu-Babovsky type, described in detail in Algorithm 1 in which $T > 0$ is the final simulation time, while P is the initial total population.

Remark 7. In line 17, $\bar{z}_N$ is an arbitrary marginal density w.r.t. the population: $\bar{z}_N(s, t) \equiv \int_{\mathcal{I}} z(i, s, t)\, \mathrm{d}i$, where z is similarly defined as (5) and should be understood as f with $\mathbf{A}$ and as g with $\mathbf{B}$.

Remark 8. More generally, in line 1, $\mathcal{S}_0$ can be sampled from an initial density $\bar{z}_0$; however, since none is known for the population distribution, it was decided to define $\mathcal{S}^0$ as a uniform vector with respect to an initial total population P.

3.2. Numerical results

3.2.1. Foreword

Firstly, all data used was taken from the Italian National Institute of Statistics (*ISTAT*) in the year 199I and limited to the Sardinia region, choice primarily inspired by [6]; in detail, [13] provided the data to build the graph and its adjacency matrix of the commuting matrix, while [12] the actual population distribution to which the results were also compared. Therefore, given the importance of Sardinia, the main data of the region are listed in Table 1 along with the fitted parameters of the Pareto and bilognormal distributions.

Table 1
Data [12, 13] and parameters of Sardinia in 1991.

N	P	β	ξ	μ_1	σ_1	μ_2	σ_2
375	$\approx 1.648\times10^6$	1.273	0.929	3.227	0.408	4.099	0.556

Remark 9. With minor, medium, and major we refer to centres with small, medium, and large degrees respectively.

Regarding the plots, their complex definitions are not necessary to understand the results shown hereafter; hence, they are omitted but still present in Appendix A.4 for the interested reader. Nonetheless, only three features have to be kept in mind: due to the stochastic nature of Algorithm 1, to have significant results $R = 100$ simulations have to be considered and later averaged in some way; moreover the time step is always $\Delta t = 0.01$ with N_t number of time steps; finally, for brevity sake, only the best fitting configuration empirically found by the authors is analyzed.

3.2.2. Size rule

A first possibility[5] consists in

$$E(s, s_*) \equiv \lambda \frac{(s_*/s)^\alpha}{1 + (s_*/s)^\alpha}, \tag{ER}_S$$

where $\lambda \in (0, 1)$ (Hypothesis 4) and $\alpha \in \mathbb{R}^+$. The logic is that the emigration rate towards the receiving city s_* is greater the larger its relative population, given by the ratio s_*/s, compared to the interacting city s.

In total, two reference configurations described by the parameters in Table 2a were studied, while the results are illustrated in Figures 4 and 5; these describe two behaviours:

1. a stable one but not distant from the initial uniform distribution, and
2. an unstable one in which *on average* the medium-sized centres depopulate.

Mind that «On average» is emphasized as in single realizations it is much more likely that the larger centres depopulate, whereas over multiple realizations at least one with a large size manifests, thus balancing the average; this is also the reason why only 10 iterations were considered instead of the usual 100 in Table 2a, given that the average tends to mask the depopulation, which is nevertheless inevitable but simply slower. In addition to Figures 5j-5l, the histograms in Figures 5a-5c also confirm this tendency in the form of a tail that elongates the distribution into negative values, corresponding to the exponents of the cities that depopulate.

More generally, it has been verified, albeit only empirically, that for every λ there exists an $\bar\alpha \in \mathbb{R}_+$ such that $\alpha < \bar\alpha$ results in a stable configuration while $\alpha \geq \bar\alpha$ yields an unstable one.

3.2.3. Size-degree rule

In order to correct $[\text{ER}]_S$, the relative degree ratio can also be introduced:

$$E_{SK}(s, i, s_*, i_*) \equiv \lambda \frac{[(s_*/s)(k_{i_*}/k_i)]^\alpha}{1 + [(s_*/s)(k_{i_*}/k_i)]^\alpha}. \tag{ER}_{SK}$$

Thus, the logic becomes that the emigration rate towards the receiving city (s_*, k_*) is higher the more populous and connected it is compared to the interacting city (s, k)

The studied reference configuration is described by the parameters in Table 2b, while the results are illustrated in Figure 6. Overall, they are better than those obtained using $[\text{ER}]_S$:

- Figures 6a, 6d and 6j illustrate that the approximation corresponds almost perfectly to the exact simulation;
- in Figures 6b and 6c the simulated distributions are fairly similar to the real ones, whereas in Figures 6e and 6f the Pareto indices are reasonably close;
- with the exception of the real case in which they coincide, in Figures 6g-6i the bilognormal follows the tail better towards the end, although it coincides with the Pareto in the first half;

[5] It is a Hill function of order α, partially inspired by [10, (2.2), p. 223, § 2, and (4.5), p. 228, § 4], widely applied in the Biomedical field.

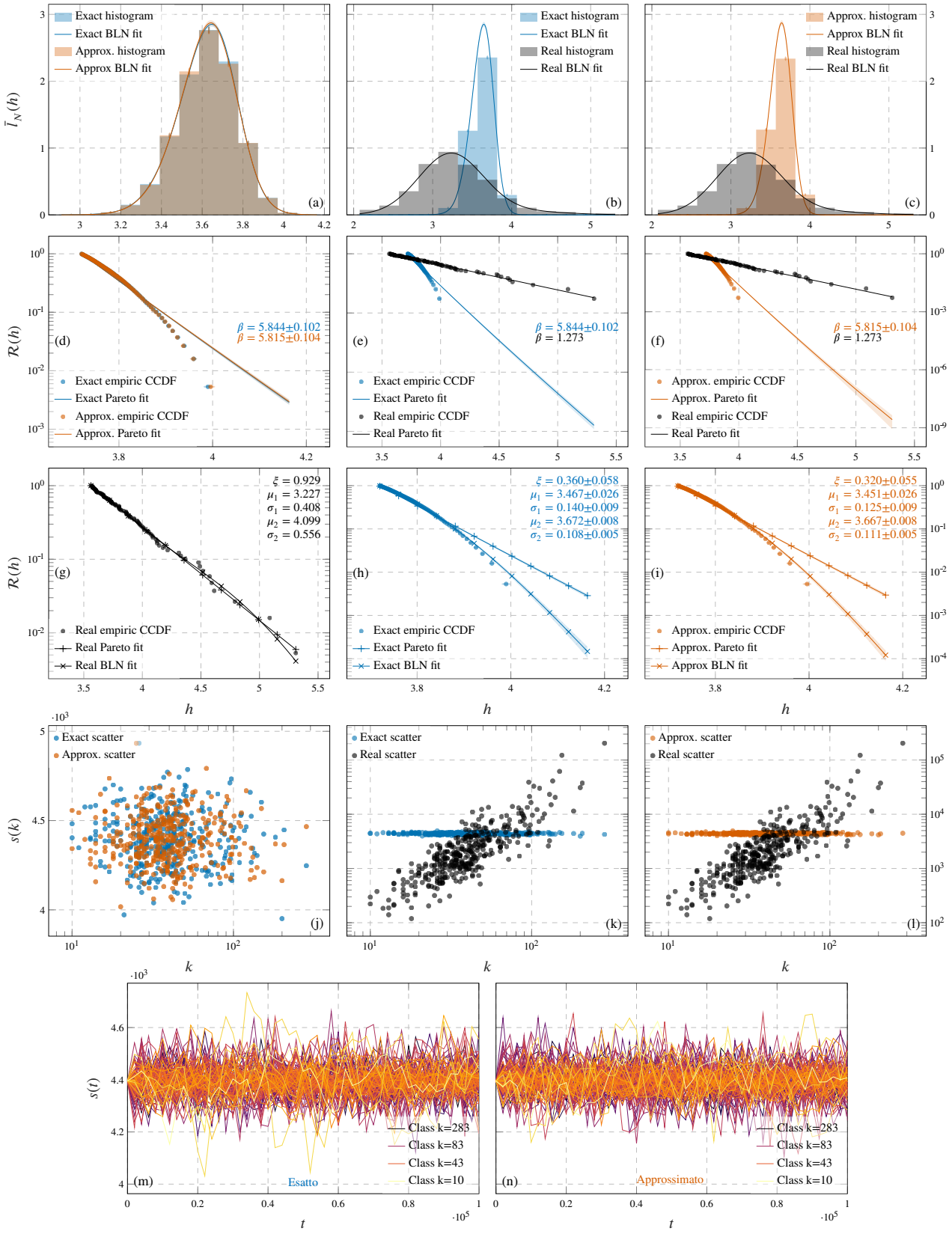


Figure 4: Study of the stable reference configuration of $[\mathrm{ER}]_S$ with parameters from Table 2a; for the explanation of the graphs, see Appendix A.4.

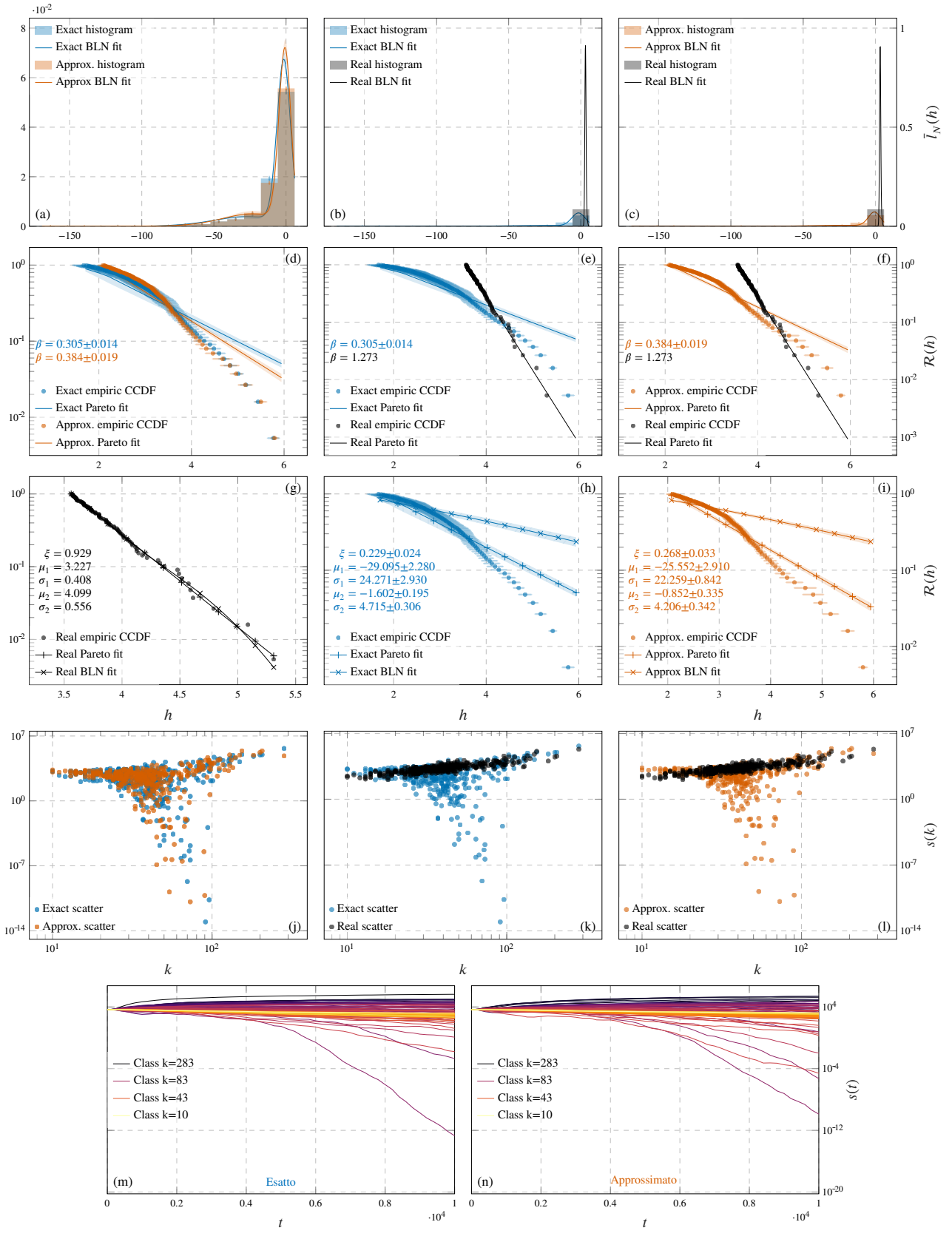


Figure 5: Study of the unstable reference configuration of $[\mathrm{ER}]_S$ with parameters from Table 2a; for the explanation of the graphs, see Appendix A.4.

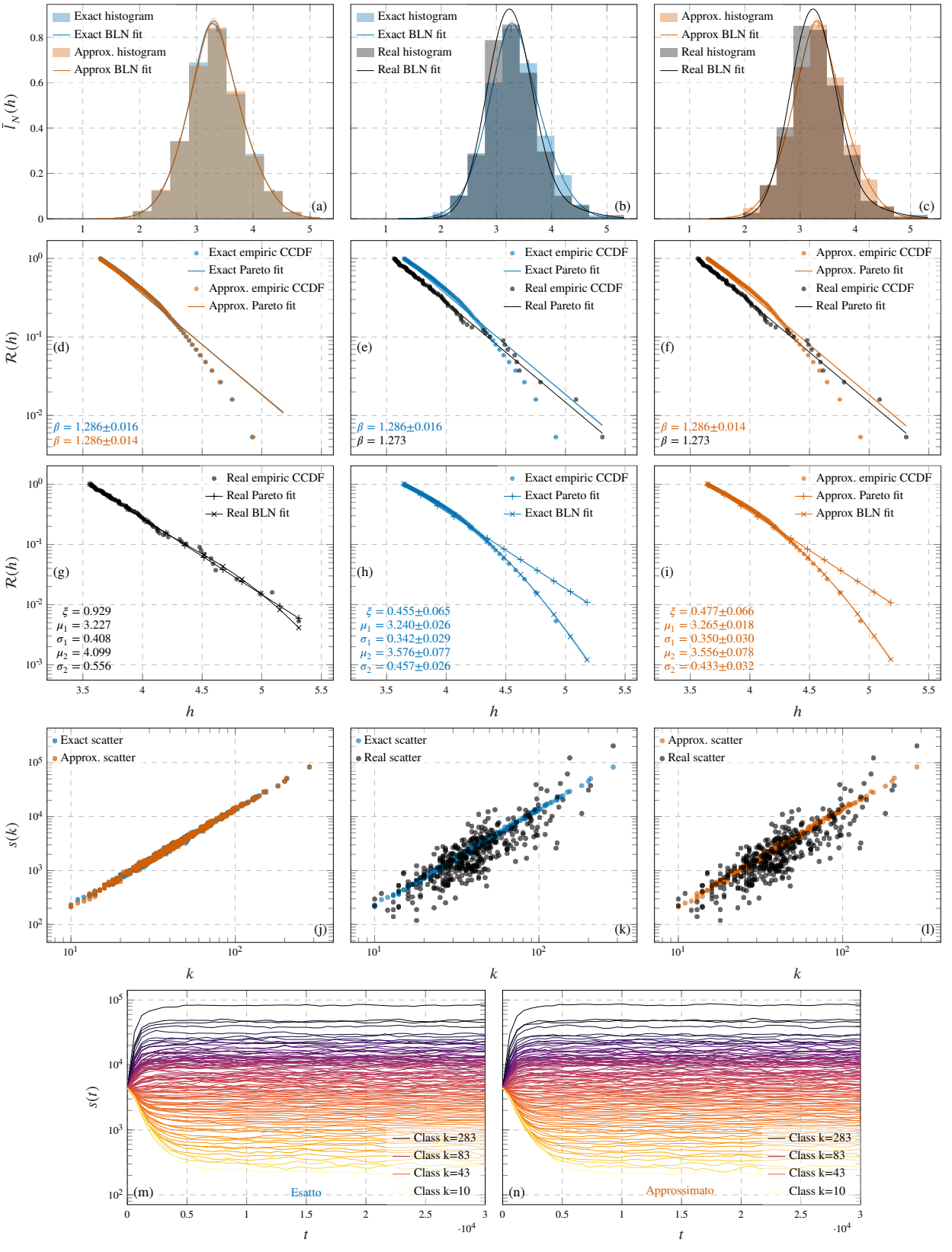


Figure 6: Study of the reference configuration of $[\mathrm{ER}]_{SK}$ with parameters from Table 2b; for the explanation of the graphs, see Appendix A.4.

Table 2
Parameters of all the reference configurations studied.

	λ	α	N_t	R
Stable	0.1	0.5	10^7	100
Unstable	0.1	1.5	10^6	10

(a) Figures 4 and 5.

λ	α	N_t
0.15	0.63	3×10^6

(b) Figure 6.

λ	α	N_t
0.18	0.44	3×10^6

(c) Figure 7.

- in Figures 6k and 6l the scatter plots present a similar positive correlation, but much tighter compared to the real one: it is almost a one-to-one correlation, which is, of course, unrealistic.

3.2.4. Size-strength rule

One may think to substitute the strengths, i.e. the total amount of commuters, in place of the degrees in $[\mathrm{ER}]_{SK}$:

$$E_{SW}(s,i,s_*,i_*) \equiv \lambda \frac{[(s_*/s)(w_{i_*}/w_i)]^\alpha}{1+[(s_*/s)(w_{i_*}/w_i)]^\alpha}, \qquad [\mathrm{ER}]_{SW}$$

meaning, in this way, that there is an emigration rate towards the receiving city (s_*, w_*) that is greater the more populous and heavily trafficked it is compared to the interacting city (s, w).

The analyzed reference configuration is described by the parameters in Table 2c, while the relative results are shown in Figure 7. They are in fact the best ones:

- Figures 7a, 7d and 7j confirm that the approximation corresponds almost perfectly to the exact simulation;
- in Figures 7b and 7c the simulated distributions are almost identical to the real one, and the relative Pareto indices in Figures 7e and 7f also correspond;
- Figures 7h and 7i do not deviate significantly from the previous configuration;
- in Figures 7k and 7l the scatter plots, besides having the same positive correlation with respect to the real one, show greater spread, correctly capturing the size differentiation for a given degree.

The only negative aspect is found in Figures 7k and 7l: the emigration rule fails to correctly predict on average the sizes of the lower portion of the real scatter plot; in other words, the prediction overestimates the size of the minor centres and slightly underestimates that of the major centres. Yet, on second thought, this is a characteristic also shared by the other case in Figures 6k and 6l, as can be noted by observing the slope of the line described by the scatter plot. We believe its cause to be found in the absence of intra-urban commuters in the strength definition which only considers inter-urban ones; this would increase only the strength of the medium and major centres, but only marginally that of the minor ones, correcting in theory the slope.

3.3. Interpretations

First of all, it is necessary to elucidate a characteristic shared by all previous emigration rules: from an interpretative point of view, they are myopic, meaning they do not see the details but only the common factors that might govern migrations. Therefore, given the arbitrariness of potential interpretations, it has been decided in this subsection not to express details, but only to highlight which factors most likely influence migration in light of the comparisons with real data in Figures 4–7.

3.3.1. Size rule - $[ER]_S$

From the results of $[\mathrm{ER}]_S$, it is clear that considering only the relative population is not sufficient to describe the size distribution among cities. The reason for its degeneration can be attributed to a binary behaviour: if $s \leq s_*$ then s_* will always be an attractive pole for s [and vice versa]; therefore, as soon as a city loses even a little population compared to all its neighbouring nodes, its fate is sealed to depopulate. With this logic, the depopulation *on average* of the medium centres can be explained: the major centres deplete the populations of the medium centres, rendering them relatively repellent poles right from the beginning of the simulation; in the short term, the medium centres depopulate to such an extent that they become repellent even with respect to the minor centres; the minor centres never interface

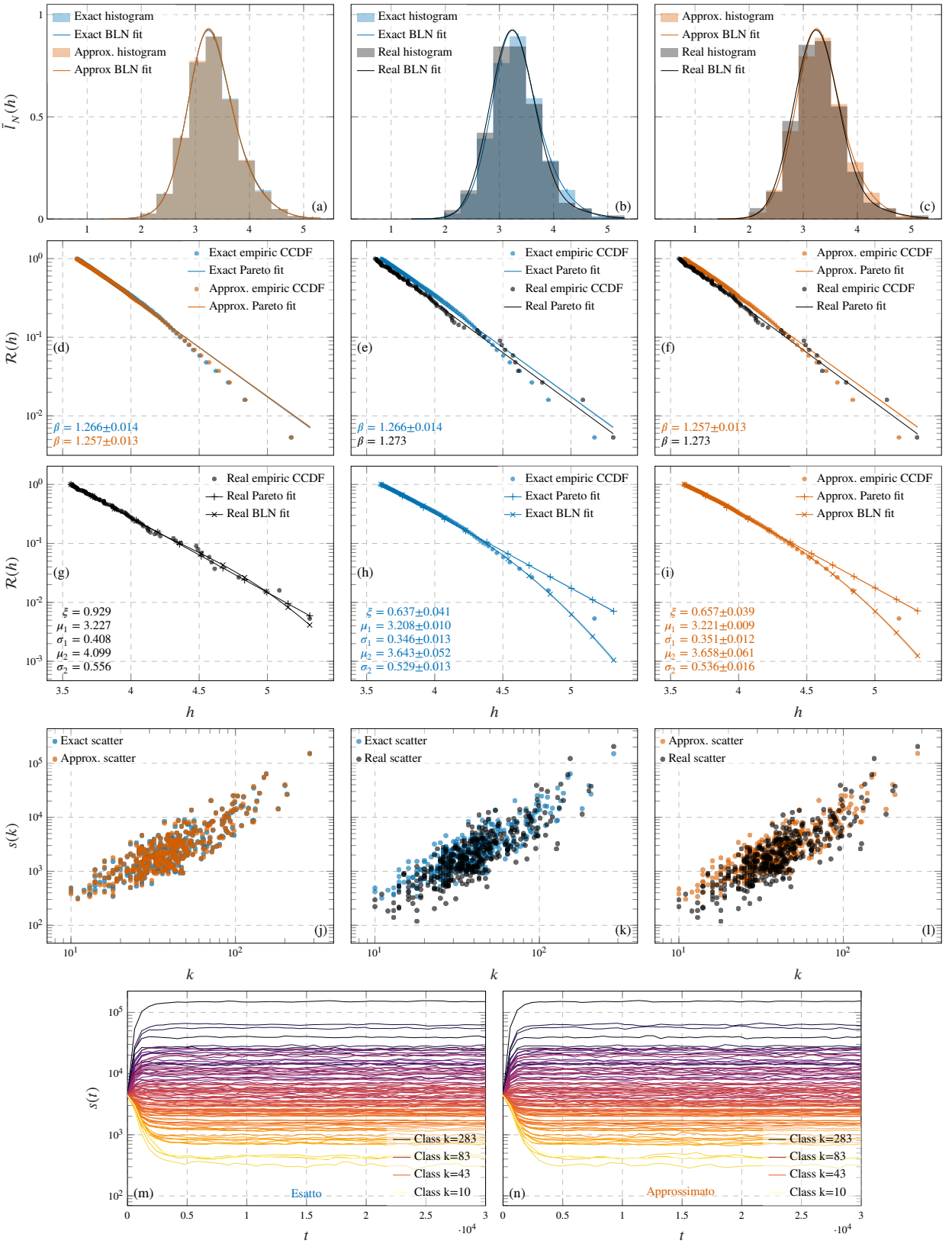


Figure 7: Study of the reference configuration of $[\mathrm{ER}]_{SW}$ with parameters from Table 2c; for the explanation of the graphs see Appendix A.4.

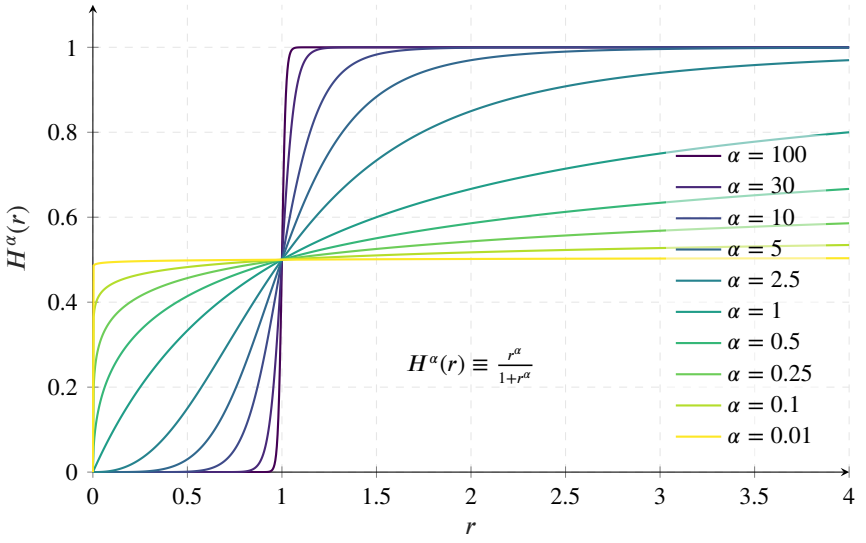


Figure 8: Hill function as the degree α varies.

directly with the major ones, therefore they do not initially depopulate like the medium ones. The result of this dynamics is shown in Figures 5k and 5l, at least in the unstable case.

Conversely, in the stable case, it is unclear what this behaviour can be attributed to: certainly α is not small enough to make the Hill function in $[\mathrm{ER}]_S$ constant (Figure 8) and equal to $\lambda/2$, especially considering that $[\mathrm{ER}]_{SW}$ in the reference configuration from Table 2c also presents a lower value of α; consequently, the only conclusion that can be drawn is that the problem lies in the emigration rule itself.

Overall, therefore, it can be deduced that migration most likely does not depend *solely* on relative population.

3.3.2. Size-degree rule - $[ER]_{SK}$

These results are decidedly more realistic; however, contrary to relative population, the introduction of the relative degree via the ratio k_{i_*}/k_i is not intuitive. In other words, one might wonder why it is more reasonable to postulate that a city with a low degree, i.e., less connected, also has a low population. The fear is that the model is being forced toward results that, although corresponding to real ones, are not realistic in their deepest logic; nevertheless, to motivate this hypothesis it is sufficient to highlight two naturally observable constraints.

V1 The first is of a physical nature induced by the spatial graph: by their nature, edges occupy spaces between cities and connect to them by taking a portion of their boundary; therefore, the only way a city can increase its degrees is to expand to have a larger boundary, but this necessarily implies an increase in population because the latter must maintain it.

V2 The second is of a social nature induced by commuter flows: greater connectivity implies a more complex transport network, which can therefore be justified if there are enough commuters using it; these, however, can be attracted to a city if it has a sufficiently high population that naturally motivates such flows.

The second constraint is indeed confirmed by Figure 9, which reports a positive correlation between degree and strength: the greater the former, the greater the latter will be, i.e., the commuter flow, from which V2 follows.

Therefore, the physical constraint justifies this type of interaction regardless, because the graph is necessarily spatial; similarly, the social constraint motivates the interaction if and only if there is a positive correlation between degree and strength[6]. All this implies the relation $s \propto k$, also confirmed by the same real dispersion in Figures 6k and 6l, which makes the introduction of k_{i_*}/k_i realistic from a modelling perspective.

It is also noteworthy to dwell on, albeit without a conclusive answer, why the emigration rule converges at all: one might wonder, in fact, why the flow minor centers → medium centers → major centers does not manifest itself.

[6]This characteristic is also confirmed for each Italian region, as well as for Italy itself.

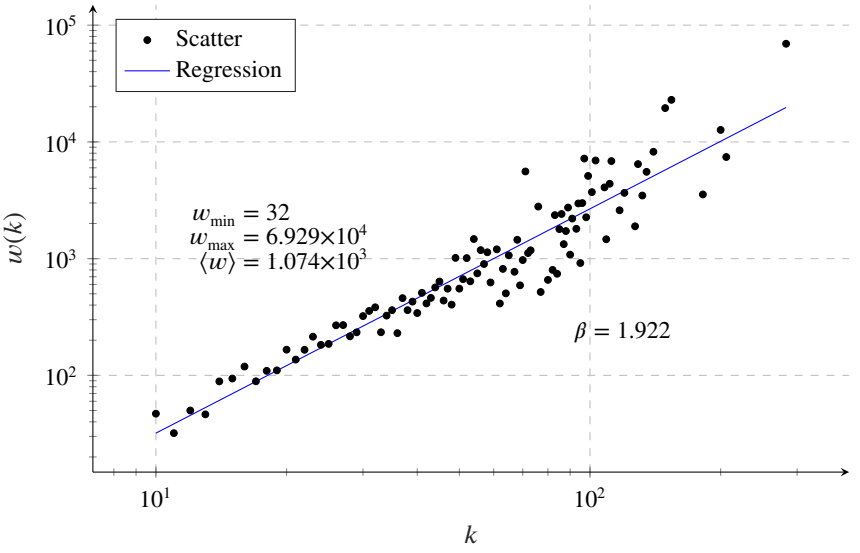


Figure 9: Strength vs degree of the Sardinia network.

The only reason for the lack of degeneration in the model can be traced back to the presence of a backflow naturally induced by $[\mathrm{ER}]_{SK}$, which balances the flow toward relatively more populated and connected centers; in other words, not only does the introduction of degrees in $[\mathrm{ER}]_S$ make medium centers attractive compared to minor ones in the short term, but it also stabilizes long-term interactions, i.e., stationary ones. Yet the exact dynamics establishing a stabilizing reverse flow are unclear, since the ratio r in the Hill function (Figure 8) should be smaller in $[\mathrm{ER}]_{SK}$ than in $[\mathrm{ER}]_S$.

In conclusion, from these results, $[\mathrm{ER}]_{SK}$ implies that migration is probably governed by *at least* two factors: the relative population and connectivity between two cities.

3.3.3. Size-strength rule - $[ER]_{SW}$

The introduction of strength is motivated by the same constraints described to motivate the introduction of degree in $[\mathrm{ER}]_S$. Reasoning this way, it is now possible to distinguish cities that have the same degree but not necessarily the same traffic; this natural distinction allows correcting the almost bijective relationship between degree and population in Figures 6k and 6l, transforming it into a more realistic positive correlation described by Figures 7k and 7l.

Overall, given that these results are the best among those seen, $[\mathrm{ER}]_{SW}$ implies that *most likely* the migration phenomenon is governed by two factors: the relative population and traffic between two cities.

3.3.4. λ and α Interpretation

In all analyzed emigration rules there are two main parameters: λ and α; to try to understand their meaning, one can proceed through two parametric studies: both take the reference configuration of $[\mathrm{ER}]_{SW}$ in Table 2c and perturb only one of the two parameters leaving the other unchanged; Table 3 summarizes what has been stated where $N_p = 5$ indicates the number of values uniformly sampled in the two intervals.

Table 3
Parametric studies in the supplementary materials.

	N_p	λ	α	N_t
Parametric study λ	5	$[0.01, 0.3]$	0.44	10^7
Parametric study α	5	0.18	$[0.3, 0.6]$	3×10^6

Before proceeding, it is necessary to keep in mind the difference between μ and the lognormal average:

Remark 10. By mass conservation (Hypothesis 5), the sample mean P/N is constant regardless of the chosen parametric pair (λ, α). However, this does *not* imply that the parameter μ in $X \sim \mathrm{Lognormal}(\mu, \sigma)$ is also constant; indeed, what is

preserved is its expected value of the form $\mathbb{E}[X] = 10^{(\mu+\sigma^2 \ln(10)/2)} = P/N$, from which it is immediately understood that if the variance σ^2 were to increase, μ would decrease and vice versa.

As can be seen in Figure 1 this behaviour is hard to perceive both in linear and logarithmic scales, while in logarithmic space everything changes since μ and σ correspond precisely to the mean and variance of the equivalent Gaussian. A similar reasoning also apply to the bilognormal fit (4), as it is a convex combination of two lognormals.

Regarding the first parametric study, varying λ, overall the following holds:

- the variance increases, broadening the distribution for increasing λ, albeit slightly;
- the Pareto index de facto remains almost unaltered, varying between the extremes by only ≈ 0.05 on average;
- the relationship between Pareto and bilognormal remains unchanged: it confirms in any case that the second follows the end of the tail better than the first, while they coincide at the beginning;
- the scatter is unaltered as λ varies: the average behaviour is the same;
- all considered cases converge with shorter convergence time for increasing λ.

It is noted that all these variations are nonetheless slight, considering that λ increases by an order of magnitude; this means that λ has little influence on the final results, except for the convergence time; However, for $\lambda > 0.3$, an unstable configuration is still reached. Regarding the second parametric study, varying α, overall the following holds:

- as λ grows, the variance increases significantly, in so decreasing the mean and broadening the distribution;
- the Pareto index varies significantly, decreasing by nearly an order of magnitude;
- the relationship in the tail between Pareto and bilognormal continues to remain unchanged;
- the dispersion initially overestimates the minor-medium centres and underestimates the major ones, to then invert this relationship with increasing α;
- all considered cases converge, with a longer convergence time for increasing α.

Contrary to the λ parametric study, the variation of the results is significant; in particular, the Pareto index β seems to largely depend on the value of α; similarly, for values of $\alpha > 0.6$, an unstable configuration is obtained. Altogether, in light of these studies, λ and α can be interpreted as follows:

- λ is the attraction: the greater its value, the more the relatively larger centers succeed in attracting, in a single interaction, high fractions of the population of the relatively smaller centers;
- α is the repulsion: the greater its value, the more the relatively larger centers are repelled by the relatively smaller ones; it is clearly inversely related to the magnitude of the stabilizing backflow.

Another point of view is the following: λ mainly influences the portion of the Hill function (Figure 8) beyond unity ($r > 1$), and thus the attraction, while α conditions more the portion below unity ($r \leq 1$), and therefore the repulsion.

4. Conclusions

All results illustrated in the previous section corroborate the node-agent perspective in an urban context to describe both the city size distribution and their Pareto index. The unit-rank approximation from Definition 2 produces results corresponding almost perfectly to those obtained exactly, confirming [16]. All interaction rules faithfully reproduce the Pareto index of the actual distribution tail, while the distribution itself was overall well reproduced with almost every considered rule. In any case, the bimodal lognormal distribution fits the tail better than the Pareto distribution, confirming [10, 9]; however, both fits present advantages and disadvantages: from an engineering perspective, the Pareto distribution is easier to handle having only one parameter, although it is simultaneously worse mathematically; conversely, the bimodal lognormal distribution is mathematically more correct but consists of 5 parameters, making any potential comparison more difficult.

Of course, the work presented here is naturally non-exhaustive and leads to multiple future developments:

1. experiment with interaction rules containing other graph theory coefficients, such as betweenness centrality or the clustering coefficient, and by adding intra-urban commuters in ($[\mathrm{ER}]_{SW}$);
2. derive the Fokker-Planck equation of $[\mathrm{EtB}]^{\mathbf{A}}_{S}$ to analytically study the stationary density, although the nonlinear and symmetric nature of the emigration rules proposed here makes this objective rather difficult to achieve;

3. abandon the simplifying hypothesis of the static network to describe a dynamic one evolving in parallel with the population, with a possible comparison against real data [if available];
4. try to reproduce the results with a European or international networks to confirm the validity of the current theory in other contexts or at larger scales.

A. Result reproducibility

A.1. Technical specifications

The code is available in this repository [19] and was written entirely in Python 3.13.7, while the simulations were executed on a machine with a Windows operating system and an Intel Core i5-7500 processor (quad-core, 3.40 GHz).

A.2. Confidence interval

Let $R \in \mathbb{N}_+$ be the number of performed simulations and $\mathbf{S} \in \mathbb{R}_+^R$ be the random vector of a city's population related to each simulation; then by virtue of Algorithm 1, the components $(S_1, S_2, \dots, S_R)$ of $\mathbf{S}$ are independent and identically distributed according to the density $\bar{z}_N$, a property that allows defining

$$\bar{S} \equiv \frac{1}{R}\sum_{r=1}^{R} S_r \quad \text{and} \quad V \equiv \frac{1}{R-1}\sum_{r=1}^{R}(S_r - \bar{S})^2,$$

as the sample mean and sample variance respectively. Then, given μ as the *true*[7] mean of the density $\bar{z}_N$, Student's T-distribution with parameter $R-1$ is written as

$$T = \frac{\bar{S} - \mu}{\sqrt{V/R}} \sim \text{Student}(R-1).$$

From this, one can define the [symmetric] confidence interval with confidence level α by setting

$$\mathbb{P}\left(-t_{R-1}^{\alpha/2} \le T \le t_{R-1}^{\alpha/2}\right) = 1-\alpha, \text{ where } t_{R-1}^{\alpha/2} \in \mathbb{R} \text{ is such that } \mathbb{P}\left(T < t_{R-1}^{\alpha/2}\right) = 1 - \frac{\alpha}{2}.$$

Expanding T and isolating the mean μ, we obtain

$$\mathbb{P}\left(\bar{S} - t_{R-1}^{\alpha/2}\sqrt{\frac{V}{R}} \le \mu \le \bar{S} + t_{R-1}^{\alpha/2}\sqrt{\frac{V}{R}}\right) = 1-\alpha, \text{ whence } \text{IC}_R^\alpha(\mathbf{S}) \equiv \left[\bar{S} - t_{R-1}^{\alpha/2}\sqrt{\frac{V}{R}}, \bar{S} + t_{R-1}^{\alpha/2}\sqrt{\frac{V}{R}}\right]$$

is the interval estimator defining the desired confidence interval, which can also be expressed more compactly as

$$\text{IC}_R^\alpha(\mathbf{S}) \equiv \bar{S} \pm t_{R-1}^{\alpha/2}\sqrt{V/R}. \tag{7}$$

The confidence interval is essentially the statistical error committed: it assesses how likely it is that the interval estimator (7) contains the parameter μ; in other words, IC_R^α measures the uncertainty of the mean estimation: given a sample $\mathbf{s}$, the wider the confidence interval, the more uncertain the sample mean $\bar{s}$ is as an estimation of the actual mean μ; conversely, the narrower it is, the more precise the estimation $\bar{s}$ is, in the sense that μ lies within a small neighbourhood of the sample mean. From this perspective, it is conceptually analogous to the precision of a measuring instrument.

Hypotesis 6 (Confidence level). In all results $\alpha = 0.05$ is chosen to have an interval with a 0.95 confidence level.

Remark 11. Naturally, the interval estimator (7) just derived holds as much for the random vector $\mathbf{S}$ as it does for its realizations[8] $\mathbf{s}$, which are precisely the result of the R simulations.

[7] That is to say, μ is *not* a random variable but the exact mean parameter of $\bar{z}_N$.

[8] In this case, (7) is referred to as an interval estimate.

A.3. Data structure

The data have the structure of a fourth-order tensor $\mathbf{s} \in \mathbb{R}_+^{N_f \times 2 \times N \times R}$ whose indices have the following meanings:

$$s_{i,r}^{n,a} \begin{cases} n = \text{time instant}, & a = \text{simulation type}, \\ i = \text{city index}, & r = \text{simulation number}. \end{cases}$$

The time instant is defined via three parameters in $\mathbb{N}_+$:

- $N_t \equiv \lfloor T/\Delta t \rfloor$ is the total number of simulated times;
- $N_s \ll N_t$ is the number of snapshots from the simulated times;
- $N_f < N_s$ is the number of reduced times from the snapshots.

Both the snapshots and the reductions are sampled at equally spaced intervals with steps $\Delta s = N_t/N_s$ and $\Delta f = N_s/N_f$ where it is assumed, for simplicity, that N_s and N_f are divisors of N_t and N_s respectively[9], and the same is assumed between Δt and T. By means of Algorithm 1 a total of N_t times are simulated with step Δt, of which N_s are saved in the snapshot tensor $\underset{\sim}{\mathbf{s}} \in \mathbb{R}_+^{N_s \times 2 \times N \times R}$:

$$\underset{\sim}{s}_{\cdot,r}^{n,a} \equiv \mathcal{S}_{\cdot,r}^{n\Delta s,a}, \quad \forall n,\ \forall a,\ \forall r,$$

where $\mathcal{S}_{\cdot,r}^{n\Delta s,a}$ is to be understood as the vector of predicted sizes at the $n\Delta s$-th step, in the r-th exact simulation if $a = 1$, or approximated simulation if $a = 2$; subsequently, $\underset{\sim}{\mathbf{s}}$ is convolved with respect to the time index:

$$s_{i,r}^{n,a} \equiv \frac{1}{\Delta f} \sum_{j=1}^{\Delta f} \underset{\sim}{s}_{i,r}^{(n-1)\Delta f + j,a}, \quad \forall n,\ \forall a,\ \forall i,\ \forall r,$$

that is to say, every Δf elements of the N_s snapshots are averaged. This mollification is necessary to make $\mathbf{s}$ less noisy than $\underset{\sim}{\mathbf{s}}$ and therefore more readable once plotted.

Overall, for the actual times, three forms are available depending on how the index n is interpreted:

$$t_n = n\Delta t, \ \ \forall n \in \{1, 2, \dots, N_t\}, \qquad t_n^s = t_n \Delta s, \ \ \forall n \in \{1, 2, \dots, N_s\}, \qquad t_n^f = t_n^s \Delta f, \ \ \forall n \in \{1, 2, \dots, N_f\},$$

respectively for the discretized, sampling, and reduced times; regardless, $t_{N_t} = t_{N_s}^s = t_{N_f}^f = T$ holds in any case.

Remark 12. The index n in $\mathbf{s}$ does not include 0 because it refers to the initial distribution, which is always equal to $\mathcal{S}^0$ from Algorithm 1; if one wishes to extend the tensor, however, it can be written as $\mathbf{s}_{\cdot,r}^{0,a} \equiv \mathcal{S}^0$, $\forall a$, $\forall r$.

Hypotesis 7. All results consider $\Delta t = 0.01$, $N_s = 1000$, $N_f = 50$, while N_t is chosen to have t_n slightly higher than the convergence time, if the results converge, but it is certainly a multiple of 10 to ensure that N_s is one of its divisors.

Remark 13. For most plots, only the distribution at the final time T is considered *without* time convolution, so that, for lightness of notation, the third-order tensor $\mathbf{s}^T \equiv \underset{\sim}{\mathbf{s}}^{N_s}$ can be improperly denoted by $\mathbf{s}^T \in \mathbb{R}_+^{2 \times N \times R}$.

A.4. Plot definitions

A.4.1. Histograms

The first plots consider the [normalized] histograms of the distribution at the final time T. We begin by considering the maximum and minimum elements of the tensor $\mathbf{s}^T$,

$$s_{\max} \equiv \max_{a,i,r} s_{i,r}^{T,a} \quad \text{and} \quad s_{\min} \equiv \min_{a,i,r} s_{i,r}^{T,a},$$

with which a common equally spaced grid can be defined to construct the histogram classes. Let N_c be the number of bins/classes of the grid; then, defining $\mathcal{B} \colon \mathbb{R}^N \to \mathbb{R}^{N_c}$ as the function that returns the [normalized] values of the histogram classes, the tensor $\mathbf{b} \in \mathbb{R}_+^{2 \times N_c \times R}$ is written as $\mathbf{b}_{\cdot,r}^a \equiv \mathcal{B}(\mathbf{s}_{\cdot,r}^{T,a})$, $\forall a$ and $\forall r$. Once the confidence intervals

$$\mathrm{IC}_R^\alpha(\mathbf{b}_c^a) = \bar{b}_c^a \pm t_{R-1}^{\alpha/2} \sqrt{v_c^a / R}, \ \ \forall a \text{ and } \forall c,$$

are evaluated, we can represent the mean histogram of the exact and approximated simulations with $\bar{\mathbf{b}}^a$, and the stochastic error on the classes with $\pm t_{R-1}^{\alpha/2} \sqrt{\mathbf{v}^a / R}$.

[9] Namely $N_t/N_s - \lfloor N_t/N_s \rfloor = 0$ and $N_s/N_f - \lfloor N_s/N_f \rfloor = 0$.

A.4.2. Bimodal Lognormal

The logic is similar to that of the histograms: let $\mathcal{L}(\cdot;\mathbf{s}^{T,a}_{\cdot,r}):\mathbb{R}_+\to\mathbb{R}_+$, be the density of a bilognormal distribution (4) fitted from the vector $\mathbf{s}^{T,a}_{\cdot,r}$, and let $\mathcal{L}(\cdot;\mathbf{s}^{T,a}):\mathbb{R}_+\to\mathbb{R}^R_+$ be the vector-valued function such that

$$\mathcal{L}(x;\mathbf{s}^{T,a})\equiv\left[\mathcal{L}(x;\mathbf{s}^{T,a}_{\cdot,1})\quad\mathcal{L}(x;\mathbf{s}^{T,a}_{\cdot,2})\quad\cdots\quad\mathcal{L}(x;\mathbf{s}^{T,a}_{\cdot,R})\right]^\top,$$

which essentially collects all the fits pointwise into a single vector. Then the confidence intervals take the form

$$\mathrm{IC}^\alpha_R(\mathcal{L}(x;\mathbf{s}^{T,a}))=\bar{\mathcal{L}}^a(x)\pm t^{\alpha/2}_{R-1}\sqrt{v^a(x)/R},\quad\forall x\in[s_{\min},s_{\max}]\text{ and }\forall a,$$

where, as before, $\bar{\mathcal{L}}^a(x)$ is the mean function while $\pm t^{\alpha/2}_{R-1}\sqrt{v^a(x)/R}$ is its stochastic uncertainty.

Remark 14. Since x is continuous, the set of confidence intervals forms a confidence band for the functions.

Remark 15. The confidence interval is applied both to the entire fit of the bilognormal and to its five parameters ξ, μ_1, σ_1, μ_2, and σ_2, of which only the form of the first is reported for brevity: letting $\boldsymbol{\xi}\in\mathbb{R}^{2\times R}$ be the matrix of all fractions obtained in the simulations, then it holds that $\mathrm{IC}^\alpha_R(\boldsymbol{\xi})=\bar{\xi}^a\pm t^{\alpha/2}_{R-1}\sqrt{v^a/R}$.

A.4.3. Empirical Complementary Cumulative Distribution Function

The empirical CCDF is defined in the tail of the population distribution, which however needs to be specified: in this case, the last quartile is chosen as the threshold, i.e., that value $s_{3/4}$ such that $\mathbb{P}(S\le s_{3/4})=3/4$, given any random variable S; in practice, this corresponds to finding that element $s_{3/4}$ of the tensor $\mathbf{s}^{T,a}_{\cdot,r}$ such that

$$\left|\left\{s^{T,a}_{i,r}\ge s_{3/4}\mid i\in\{1,2,\dots,N\}\right\}\right|=\left\lceil\frac{N}{4}\right\rceil,\quad\forall a\text{ and }\forall r,$$

which in words means that *at least* $1/4$ of all the values of $\mathbf{s}^{T,a}_{\cdot,r}$ are greater than or equal to $s_{3/4}$. Then the interval $[s_{3/4},s_{\max}]$ characterizes the tail.

Accordingly, the empirical CCDF, $\mathcal{R}_{\mathcal{E}}(\cdot;\mathbf{s}^{T,a}_{\cdot,r}):\mathbb{R}_+\to\mathbb{R}_+$, which approximates $\mathbb{P}(S>x)$, is written as

$$\mathcal{R}_{\mathcal{E}}(x;\mathbf{s}^{T,a}_{\cdot,r})\equiv1-\frac{1}{N}\sum_{i=1}^{N}\chi_{[0,x]}\left(s^{T,a}_{i,r}\right),\quad\forall a\text{ and }\forall r,\qquad\text{where }\chi_{[0,x]}(w)\equiv\begin{cases}1&\text{if }w\in[0,x],\\0&\text{otherwise.}\end{cases}\tag{8}$$

is the indicator function. However, since (8) must be evaluated for $x\in[s_{3/4},s_{\max}]$, and specifically on the tensor $\mathbf{p}\in\mathbb{R}^{2\times N_{1/4}\times R}$ of $\mathbf{s}$ restricted to the tail $p^a_{i,r}\equiv s^{T,a}_{N-N_{1/4}+i,r}$, $\forall a$, $\forall i\in\{1,2,\dots,N_{1/4}\}$ and $\forall r$ with $N_{1/4}\equiv\lceil N/4\rceil$, the following rescaling holds:

$$\begin{aligned}\mathcal{R}_{\mathcal{E}}(x;\mathbf{s}^{T,a}_{\cdot,r})&=1-\frac{N-N_{1/4}}{N}-\frac{1}{N}\sum_{i=1}^{N_{1/4}}\chi_{[0,x]}\left(s^{T,a}_{N-N_{1/4}+i,r}\right)\\&=\frac{N_{1/4}}{N}\left[1-\frac{1}{N_{1/4}}\sum_{i=1}^{N_{1/4}}\chi_{[0,x]}\left(p^{T,a}_{i,r}\right)\right]=\frac{N_{1/4}}{N}\,\mathcal{R}_{\mathcal{E}}(x;\mathbf{p}^a_{\cdot,r}),\quad\forall a\text{ and }\forall r.\end{aligned}$$

That is to say, $\mathcal{R}_{\mathcal{E}}(x;\mathbf{p}^a_{\cdot,r})$ is equal to $\mathcal{R}_{\mathcal{E}}(x;\mathbf{s}^{T,a}_{\cdot,r})$ up to a scaling factor $N_{1/4}/N$. At any rate, the Pareto index is invariant with respect to rescalings:

Remark 16. In logarithmic scale, (2) is written as $\log(\mathcal{R}(s))\approx-\beta\log(s)+\log(c)$, which is equivalent to a straight line with slope β and intercept $\log(c)$; it follows that $\log(\mathcal{R}'(s))=\log(\mathcal{R}(s))-\log(c)\approx-\beta\log(s)$, with $\mathcal{R}'(s)\equiv\mathcal{R}(s)/c$, for which the Pareto index β remains unchanged.

From this observation, one can therefore choose $\mathcal{R}_{\mathcal{E}}(x;\mathbf{s}^{T,a}_{\cdot,r})$ in logarithmic scale for greater readability, precisely because what is of greatest interest is not the values themselves but rather their trend encapsulated in the parameter

β. The use of this scale, however, introduces a minor issue: the last element $p^a_{N_{1/4},r}$ has a zero ordinate and is thus not visible; to obviate the problem, the empirical CCDF can be translated in linear scale:

$$\mathcal{R}_{\mathcal{E}}(x;\mathbf{p}^a_{\cdot,r}) + \frac{1}{2N_{1/4}}, \quad \forall a \text{ and } \forall r,$$

which corresponds in logarithmic scale to a non-uniform translation of the values, which are therefore distorted, albeit slightly since $1/2N_{1/4}$ is a small quantity; for this reason, this modification is valid *only graphically* and not for the computation of the Pareto index, which is always retrieved from the tensor $\mathbf{p}$.

Consequently, as a whole, the empirical CCDF is depicted in logarithmic scale by the scatter plot of the pairs

$$\left(\frac{N}{N_{1/4}}\,\mathcal{R}_{\mathcal{E}}(\bar{p}^a_i;\mathbf{s}^{T,a}_{\cdot,r}) + \frac{1}{2N_{1/4}}, \bar{p}^a_i\right) = \left(\mathcal{R}_{\mathcal{E}}(\bar{p}^a_i;\mathbf{p}^a_{\cdot,r}) + \frac{1}{2N_{1/4}}, \bar{p}^a_i\right), \quad \forall a \text{ and } \forall i,$$

where $\bar{p}^a_i$ comes from the confidence intervals

$$\mathrm{IC}^{\alpha}_R(\mathbf{p}^a_i) = \bar{p}^a_i \pm t^{\alpha/2}_{R-1}\sqrt{v^a_i/R}, \quad \forall a \text{ and } \forall i.$$

Remark 17. Note that, contrary to the previous cases, the confidence interval is horizontal and not vertical; indeed, the image of the empirical CCDF is invariant with respect to the simulations, which only modify the populations of the largest hubs, i.e., the abscissa of the empirical CCDF.

A.4.4. Pareto CCDF and Relative Index

Regarding the fitting of the Pareto distribution, the reasoning is analogous to that seen for the bilognormal: let $\mathcal{R}_{\mathcal{P}}(\cdot;\mathbf{p}^a_{\cdot,r}) : \mathbb{R}_+ \to \mathbb{R}_+$, $\forall a$ and $\forall r$, be the CCDF of the Pareto distribution density, analogous to (2), fitted from the vector $\mathbf{p}^a_{\cdot,r}$, and let $\mathcal{R}_{\mathcal{P}}(\cdot;\mathbf{p}^a) : \mathbb{R}_+ \to \mathbb{R}_+$ be the vector-valued function such that

$$\mathcal{R}_{\mathcal{P}}(x;\mathbf{p}^a) \equiv \begin{bmatrix}\mathcal{R}_{\mathcal{P}}(x;\mathbf{p}^a_{\cdot,1}) & \mathcal{R}_{\mathcal{P}}(x;\mathbf{p}^a_{\cdot,2}) & \cdots & \mathcal{R}_{\mathcal{P}}(x;\mathbf{p}^a_{\cdot,R})\end{bmatrix}^{\top};$$

then the confidence intervals take the form

$$\mathrm{IC}^{\alpha}_R\left(\frac{N}{N_{1/4}}\,\mathcal{R}_{\mathcal{P}}(x;\mathbf{p}^a)\right) = \bar{\mathcal{R}}^a_{\mathcal{P}}(x) \pm t^{\alpha/2}_{R-1}\sqrt{\frac{v^a(x)}{R}}, \quad \forall x \in [s_{3/4}, s_{\max}] \text{ and } \forall a,$$

where $N/N_{1/4}$ is the rescaling factor described above and is necessary to compare the fit with the empirical CCDF.

Remark 18. As in Remark 15, the confidence interval is applied both to the Pareto CCDF and to its parameter β: letting $\boldsymbol{\beta} \in \mathbb{R}^{2\times R}$ be the matrix of all Pareto indices obtained in the simulations, it then holds that

$$\mathrm{IC}^{\alpha}_R(\boldsymbol{\beta}) = \bar{\beta}^a \pm t^{\alpha/2}_{R-1}\sqrt{v^a/R}, \quad \forall a.$$

A.4.5. Bimodal Lognormal CCDF

The bilognormal is necessary not only for the entire distribution but also for the tail: let

$$\mathcal{R}_{\mathcal{L}}(\cdot;\mathbf{p}^a_{\cdot,r}),\ \mathcal{R}_{\mathcal{B}}(\cdot;\mathbf{p}^a_{\cdot,r}) : \mathbb{R}_+ \to \mathbb{R}_+, \quad \forall a \text{ and } \forall r,$$

be the CCDFs of the lognormal and bilognormal distribution densities respectively, fitted to the vector $\mathbf{p}^a_{\cdot,r}$; then by linearity of the CCDF, induced by the linearity of the integral, from (4) it holds that

$$\mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a_{\cdot,r}) = \xi\,\mathcal{R}_{\mathcal{L}}(x;\mathbf{p}^a_{\cdot,r}) + (1-\xi)\,\mathcal{R}_{\mathcal{L}}(x;\mathbf{p}^a_{\cdot,r}), \quad \forall a \text{ and } \forall r.$$

Therefore, as before, we can define $\mathcal{R}_{\mathcal{B}}(\cdot;\mathbf{p}^a) : \mathbb{R}_+ \to \mathbb{R}_+$ as the vector-valued function such that

$$\mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a) \equiv \begin{bmatrix}\mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a_{\cdot,1}) & \mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a_{\cdot,2}) & \cdots & \mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a_{\cdot,R})\end{bmatrix}^{\top},$$

which leads to the confidence intervals

$$\mathrm{IC}^{\alpha}_R\left(\frac{N}{N_{1/4}}\,\mathcal{R}_{\mathcal{B}}(x;\mathbf{p}^a)\right) = \bar{\mathcal{R}}^a_{\mathcal{B}}(x) \pm t^{\alpha/2}_{R-1}\sqrt{\frac{v^a(x)}{R}}, \quad \forall x \in [s_{3/4}, s_{\max}] \text{ and } \forall a.$$

A.4.6. Data Transformation

As shown in Figure 1(c), if some data $\mathbf{s}$ are distributed according to a lognormal, then their reformulation $\mathbf{h} \in \mathbb{R}_+^{N_f \times 2 \times N \times R}$ in logarithmic space $h_{i,r}^{n,a} \equiv \log(s_{i,r}^{n,a})$, $\forall n$, $\forall a$, $\forall i$, $\forall r$, is normally distributed; the advantage of using $\mathbf{h}$ instead of $\mathbf{s}$ lies in the easier interpretability of the data, especially when parameters vary, and thus in parametric studies like those shown in the supplementary material.

This transformation is therefore applied to the plots defined so far, which change in the following way: first, the histograms are computed as before but with the reformulated data, so that the class values becomes $\mathbf{b}_{\cdot,r}^{a} \equiv \mathcal{B}(\mathbf{h}_{\cdot,r}^{T,a})$, $\forall a$ and $\forall r$, and similarly for the confidence intervals. It is equally important to point out that in the space $h \equiv \log(s)$, the density $\bar{f}_N(s) \equiv \bar{f}_N(s,T)$ also transforms[10] while preserving the total probability; this translates mathematically into

$$\mathrm{d}P = \bar{f}_N(s)\mathrm{d}s = \bar{l}_N(h)\mathrm{d}h = \bar{l}_N(h)\frac{\mathrm{d}s}{s\ln(10)} \implies \bar{l}_N(h) = 10^h \ln(10)\bar{f}_N(10^h),$$

where $\mathrm{d}P$ is the probability variation relative to the variations $\mathrm{d}s/\mathrm{d}h$ at generic points s/h, while $\bar{l}_N(h)$ is the transformation of $\bar{f}_N(s)$ into the new space h.

On the other hand, the empirical CCDFs, both Pareto or bilognormal, remain unchanged and are simply remapped from the interval $[s_{\min}, s_{\max}]$ to $[h_{\min}, h_{\max}]$, with $h_{\min} = \log(s_{\min})$ and $h_{\max} = \log(s_{\max})$. This follows from the fact that the logarithm is a strictly increasing function, thereby preserving the order of the transformed data:

$$\mathbb{P}(X < x) = \mathbb{P}(\log(X) < \log(x)), \quad \forall x \in \mathbb{R}_+,$$

given a generic random variable $X \in \mathbb{R}_+$, it holds that:

Remark 19. From Remark 16, transforming s into h, the straight line becomes $\log(\mathcal{R}'(h)) \approx -\beta h$, which is the same relation but in semilogarithmic scale, hence this is the scale used in all previous CCDF plots.

A.4.7. Evolution of the Mean Size

Expressing the tensor of the mean sizes with respect to the nodes as $\bar{\mathbf{s}} \in \mathbb{R}_+^{N_f \times 2 \times R}$

$$\bar{s}_r^{n,a} \equiv \frac{1}{N}\sum_{i=1}^{N} s_{i,r}^{n,a}, \quad \forall n, \ \forall a \text{ and } \forall r,$$

where the confidence intervals of $\bar{\mathbf{s}}$ are

$$\mathrm{IC}_R^{\alpha}(\bar{\mathbf{s}}^{n,a}) = \bar{\bar{s}}^{n,a} \pm t_{R-1}^{\alpha/2}\sqrt{v^{n,a}/R}, \quad \forall n \text{ and } \forall a,$$

then the trend of the total mean size is depicted by linearly interpolating the pairs $\left(t_n^f, \bar{\bar{s}}^{n,a}\right)$, $\forall n$ and $\forall a$, at contiguous time instants with a confidence band of $\pm t_{R-1}^{\alpha/2}\sqrt{v^{n,a}/R}$.

A.4.8. Mean Sizes versus Degrees

Let us now denote by $\bar{\mathbf{s}} \in \mathbb{R}_+^{2\times R}$ the tensor of the mean sizes with respect to the simulations

$$\bar{s}_i^a \equiv \frac{1}{R}\sum_{r=1}^{R} s_{i,r}^{T,a}, \quad \forall a \text{ and } \forall i;$$

thus, the pairs $\left(k_i, \bar{s}_i^a\right)$, $\forall a$ and $\forall i$, defines the scatter plot linking the mean size of a node across all simulations to its degree.

Remark 20. In this case, confidence intervals are not drawn because the plot is already very dense and adding further vertical/horizontal bars would make it difficult to read.

[10] More specifically, as already mentioned at the beginning, it tends to a Gaussian density.

A.4.9. Evolutions of the Mean Sizes of Degree Classes

Let us denote by $\bar{s} \in \mathbb{R}^{N_f \times 2 \times |\mathcal{K}|}$ the tensor

$$\bar{s}_k^{n,a} \equiv \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} \frac{1}{R} \sum_{r=1}^{R} s_{i,r}^{n,a}, \quad \forall n, \ \forall a \text{ and } \forall k,$$

where $\mathcal{K} \subset \{1, 2, \dots, N\}$ is the set of distinct degrees of $\mathbf{k}$, while $\mathcal{I}_k \equiv \{i \in \mathcal{I} \mid k_i = k\} \subseteq \mathcal{I}$ represents the set of indices with degree k; therefore, by linearly interpolating the pairs $\left(t_n^f, \bar{s}_k^{n,a}\right)$, $\forall n$, $\forall a$ and $\forall k$, at contiguous time instants, one can represent the time evolution of the mean population of the k-th class across all simulations. This plot is then used to ensure that the system is no longer in the transient phase and has therefore reached a steady state.

We conclude by noting that confidence intervals are not drawn for the same reason clarified in Remark 20.

CRediT authorship contribution statement

Valerio Taralli: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - Original Draft. **Andrea Tosin:** Supervision.

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